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Autonomous Institution Affiliated to Visvesvaraya Technological University, Belagavi
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DEPARTMENT OF MATHEMATICS

Course: Fundamentals of Linear Algebra, Calculus and Differential Equations	**TEST I**	**Maximum marks: 50**
Course code: MAT211TB	**First semester 2024-2025 Physics Cycle Branch: AS, CH, ME, IEm**	**Time: 10 AM to 11:30 AM Date: 04 /11/2024**

SOLUTION AND SCHEME

Q.No	Solution and Scheme	M
1a	$[A]=\begin{bmatrix}2&3&1&5&4\\0&2&2&3&4\\4&8&4&13&\alpha\end{bmatrix}\overset{R_3-2R_1,}{\sim}\begin{bmatrix}2&3&1&5&4\\0&2&2&3&4\\0&2&2&3&\alpha-8\end{bmatrix}\overset{R_3-R_2,}{\sim}\begin{bmatrix}2&3&1&5&4\\0&2&2&3&4\\0&0&0&0&\alpha-12\end{bmatrix}$ Rank of matrix A is 2 if, and only if $\alpha=12$.	1+2+1
1b	$x=\frac{1}{8}(20+3y-2z),\ y=\frac{1}{11}(33-4x+z),\ z=\frac{1}{12}(35-6x-3y)$ **Iteration** | **x** | **y** | **z** 1 | 2.5 | 2.0909 | 1.1439 2 | 2.9981 | 2.0138 | 0.9142 3 | 3.0266 | 1.9825 | 0.9077 4 | 3.0165 | 1.9856 | 0.9120	2+1+1 +1+1
2	$[A:B]=\begin{bmatrix}2&2&1&0&:&7\\-1&2&0&0&:&-2\\-1&0&0&2&:&0\\-3&1&2&0&:&-3\end{bmatrix}\overset{R_2\leftrightarrow R_1}{\sim}\begin{bmatrix}-1&2&0&0&:&-2\\2&2&1&0&:&7\\-1&0&0&2&:&0\\-3&1&2&0&:&-3\end{bmatrix}\overset{R_2+2R_1,R_3-R_1,R_4-3R_1}{\sim}\begin{bmatrix}-1&2&0&0&:&-2\\0&6&1&0&:&3\\0&-2&0&2&:&2\\0&-5&2&0&:&3\end{bmatrix}$ $\overset{3R_3+R_2,6R_4+5R_2}{\sim}\begin{bmatrix}-1&2&0&0&:&-2\\0&6&1&0&:&3\\0&0&1&6&:&9\\0&0&17&0&:&33\end{bmatrix}\overset{R_4-17R_3}{\sim}\begin{bmatrix}-1&2&0&0&:&-2\\0&6&1&0&:&3\\0&0&1&6&:&9\\0&0&0&-102&:&-120\end{bmatrix}$ $x_1=2.353, x_2=0.1765, x_3=1.9412, x_4=1.1764$	1+5+4
3	$\begin{bmatrix}1&2&3&:&6\\1&3&5&:&9\\2&5&a&:&b\end{bmatrix}\overset{R_2-R_1,R_3-2R_1}{\sim}\begin{bmatrix}1&2&3&:&6\\0&1&2&:&3\\0&1&a-6&:&b-12\end{bmatrix}\overset{R_3-R_2}{\sim}\begin{bmatrix}1&2&3&:&6\\0&1&2&:&3\\0&0&a-8&:&b-15\end{bmatrix}$ (i) Unique solution if $a\neq 8, b$ is any value. $x=\frac{b-15}{a-8}, y=\frac{3a-2b+6}{(a-8)}, z=\frac{b-15}{a-8}$ (ii) Infinitely many solution if $a=8$ and $b=15$. $x=k, y=3-2k, z=k$ where k is arbitrary. (iii) No solution if $a=8,\ b\neq 15$.	1+3+2 +2+2

4	$\begin{bmatrix}2&1&2&1&0&0\\2&2&1&0&1&0\\1&2&2&0&0&1\end{bmatrix} \overset{R_2-R_1,\,2R_3-R_1}{\sim} \begin{bmatrix}2&1&2&1&0&0\\0&1&-1&-1&1&0\\0&3&2&-1&0&2\end{bmatrix} \overset{R_1-R_2,\;R_3-3R_2}{\sim} \begin{bmatrix}2&0&3&2&-1&0\\0&1&-1&-1&1&0\\0&0&5&2&-3&2\end{bmatrix}$ $\overset{5R_1-3R_3,\;5R_2+R_3}{\sim}\begin{bmatrix}10&0&0&4&4&-6\\0&5&0&-3&2&2\\0&0&5&2&-3&2\end{bmatrix} \overset{\frac{R_1}{10},\;\frac{R_2}{5},\frac{R_3}{5}}{\sim}\begin{bmatrix}1&0&0&\frac{2}{5}&\frac{2}{5}&\frac{-3}{5}\\0&1&0&\frac{-3}{5}&\frac{2}{5}&\frac{2}{5}\\0&0&1&\frac{2}{5}&-\frac{3}{5}&\frac{2}{5}\end{bmatrix}$ Therefore, $A^{-1}=\frac{1}{5}\begin{bmatrix}2&2&-3\\-3&2&2\\2&-3&2\end{bmatrix}$ $X=A^{-1}B=1,2,3.$	1+2+2 +2+1 +2
5a	$AX^{(0)}=12\begin{bmatrix}0.9167\\0.25\\1\end{bmatrix}, AX^{(1)}=12.083\begin{bmatrix}0.8828\\0.4413\\1\end{bmatrix}, AX^{(2)}=12.207\begin{bmatrix}0.8774\\0.5881\\1\end{bmatrix}$ $AX^{(3)}=12.3432\begin{bmatrix}0.8872\\0.6997\\1\end{bmatrix}, \quad AX^{(4)}=12.474\begin{bmatrix}0.9036\\0.7833\\1\end{bmatrix}$ Largest eigenvalue is 12.474, corresponding eigenvector $=\begin{bmatrix}0.9036\\0.7833\\1\end{bmatrix}$	5+1
5b	Characteristic equation$(\lambda^2-13\lambda+30)=0$ Eigenvalues: 3, 10 Eigenvectors: $\begin{bmatrix}-3k_1\\k_1\end{bmatrix}, \begin{bmatrix}k_2\\2k_2\end{bmatrix}$	1+1+2

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